

DAPO Make long-CoT RL learn effectively

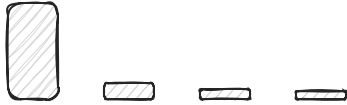
Exploration / useful groups / token balance / length

Long-reasoning bottlenecks

DAPO design responses

Entropy collapse

 Early deterministic policy



Few alternatives survive;
rare tokens gain little probability.

Clip-Higher

 Raise the upper clip only



More room for exploration;
keep the lower clip conservative.

Vanishing useful groups

    same reward
    same reward

No within-group reward contrast;
the effective training batch shrinks.

Dynamic sampling

     retain

 Resample until the batch is full

Keep mixed-correctness groups;
discard all-correct / all-wrong.

Diluted long-response tokens

Equal weight per response



Each token in a longer response
contributes less to the mean loss.

Token-level loss

Equal weight per valid token



Long reasoning retains influence;
undesired long patterns count too.

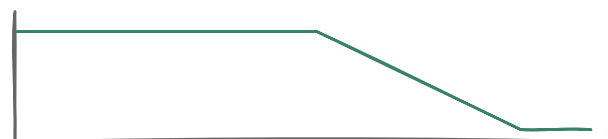
Noisy truncation penalties

 Reasoning 



A hard cutoff can punish sound
reasoning and inject reward noise.

Overlong reward shaping



No penalty Soft ramp

Shape rewards; mask truncated loss.

 Train useful, diverse reasoning signals at token granularity.